QC810: Empirical Barren Plateau Check

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Abstract

Variational quantum circuits often suffer from barren plateaus, where gradients vanish exponentially and training becomes ineffective [1]. In this work, we empirically assess the trainability of the QuantumValueNetwork (QVN) by measuring the gradient variance $\mathrm{Var}[\partial C/\partial\theta]$ across randomly sampled parameter initializations. We analyse how gradient variance scales with the number of qubits and circuit depth, identifying a maximum safe depth beyond which gradients degrade. We further compare random and identity-block initialization strategies [2], examine the impact of local versus global observables [3], and study different entanglement topologies (none, nearest-neighbour, and all-to-all). The results provide practical insights into how architectural and initialization choices influence gradient behaviour, offering guidelines for designing scalable and trainable quantum models.

1 Proposed Architectural Choices for Barren Plateau Mitigation

1.1 Local Observable

We employ local observables to restrict measurement support, preventing exponential gradient concentration [3].

1.2 Identity-Block Initialization

We use identity-block initialization to keep the circuit close to the identity at the start of training, preserving non-vanishing gradients at initialization [2].

1.3 Fixed Entangling Preamble

A fixed entangling preamble introduces structured correlations without contributing to trainable depth.

1.4 Shallow Circuit Depth

We enforce $L \leq \lfloor \log_2 n \rfloor$ (e.g., for $n = 8$, $L \leq 3$) to maintain polynomial trainability guarantees, as established by Cerezo et al. [3].

The full QVN circuit incorporating all four mitigations is shown in Figure 1.

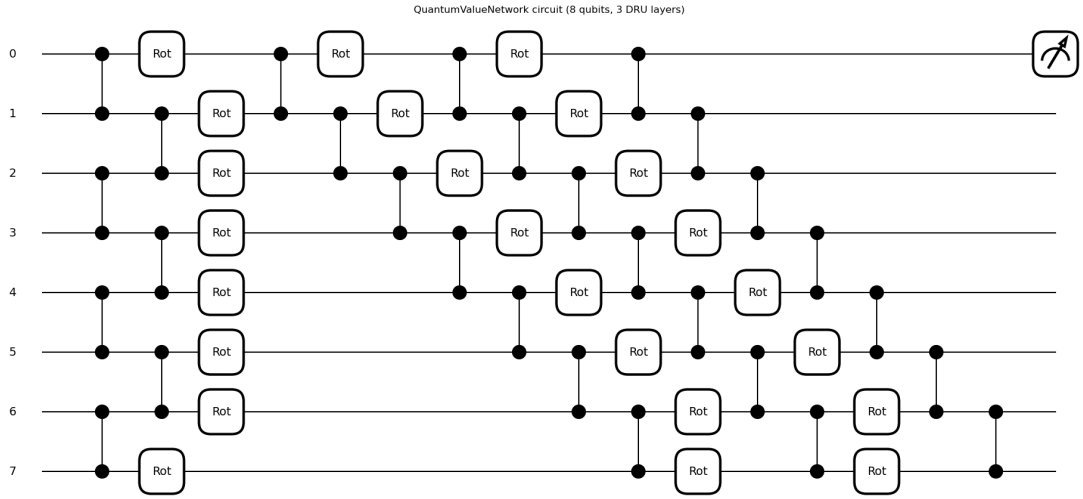

Figure 1: Proposed QuantumValueNetwork (QVN) architecture incorporating barren plateau mitigation strategies: data encoding, fixed entangling preamble, shallow variational layers initialized via identity blocks [2], and measurement with a local observable [3].

2 Gradient Variance Pipeline

For each of n_{samples} random parameter initializations, we evaluate $\nabla_\theta \langle Z_0 \rangle$ on the circuit at a freshly encoded input and record the gradient at a single fixed parameter, $\theta_{0,0,1}$ (the R_Y angle of the first Rot gate on qubit 0 in layer 0). Circuits are implemented using PennyLane [4].
The reported quantity is:

$$\mathrm{Var}[\partial C / \partial \theta_{0,0,1}] \,,$$

i.e., the variance of this parameter's gradient across random initializations, which is the canonical barren-plateau indicator used in [1, 2, 3].
We track index $(0, 0, 1)$ (not $(0, 0, 0)$) because $\mathrm{qml.Rot}(a, b, c) = R_Z(c) R_Y(b) R_Z(a)$ [4], and the first angle corresponds to an R_Z acting on a $|0\rangle$-aligned state, contributing only a global phase and yielding an identically zero gradient; the R_Y component instead produces a non-trivial, informative gradient.

3 Variance as a Function of Number of Qubits

We evaluate $\mathrm{Var}[\partial C / \partial \theta_{0,0,1}]$ as the number of qubits n increases, keeping circuit depth fixed at $L = 1$ and using random initialization. Under a local observable and shallow depth, Cerezo et al. [3] prove that gradient variance decays at most polynomially with n; we verify this empirically in Figure 2.

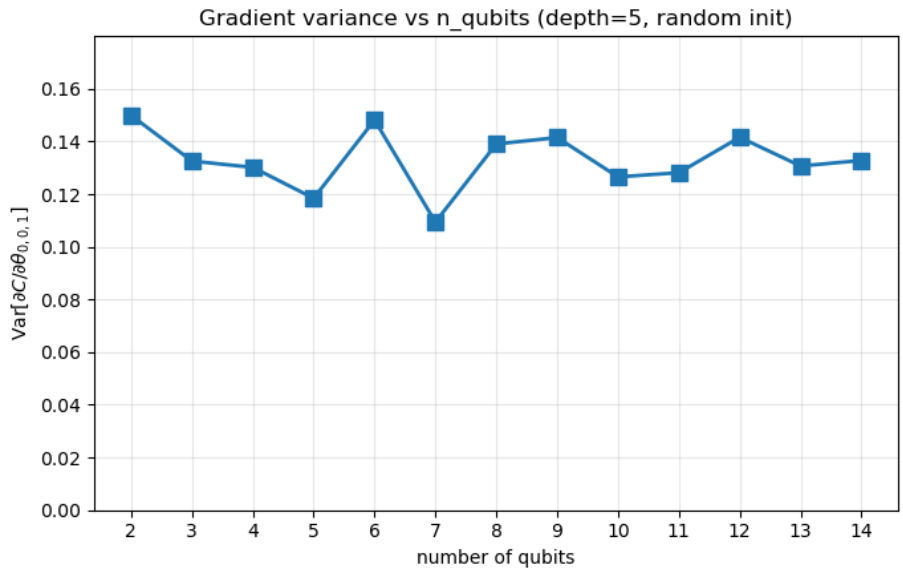

Figure 2: Gradient variance as a function of the number of qubits ($L = 1$, random init, local Z_0). The variance remains approximately constant across $n \in [2, 14]$, indicating the absence of exponential decay and hence no barren plateau in this regime, consistent with [3].

As seen in Figure 2, the variance remains $\mathcal{O}(10^{-1})$ across $n \in [2, 14]$, with a near-zero log-linear slope, indicating no exponential suppression of gradients with system size.

4 Variance as a Function of Circuit Depth (Maximum Safe Depth)

We evaluate $\mathrm{Var}[\partial C/\partial\theta_{0,0,1}]$ as a function of circuit depth L for a fixed system size ($n = 8$ qubits) under random initialization. McClean et al. [1] showed that deep random circuits converge to unitary 2-designs, producing exponentially vanishing gradients; we test whether this regime is reached within $L \in [1, 15]$, as shown in Figure 3.

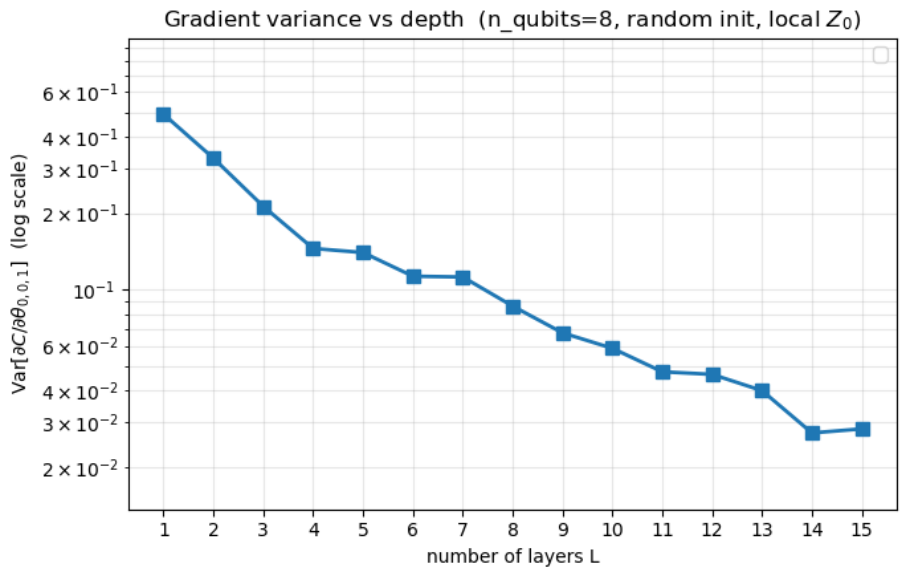

Figure 3: Gradient variance vs circuit depth (log-scale) for $n = 8$ qubits under random initialization. The variance decays with depth but does not exhibit exponential suppression, indicating a polynomial (trainable) regime consistent with [3].

As seen in Figure 3, the variance decreases gradually from $\mathcal{O}(10^{-1})$ to $\mathcal{O}(10^{-2})$ as L increases from 1 to 15, with a mildly negative log-linear slope, indicating polynomial—not exponential—decay. We identify a *maximum safe depth* around $L3 \approx \lfloor \log_2 n \rfloor$, beyond which gradients begin to diminish more noticeably, although no sharp barren plateau transition is observed.

5 Local vs Global Observable

Motivated by the theoretical results of Cerezo et al. [3], we compare gradient variance under a local observable (Z_0) versus a global observable ($Z_0 \otimes \cdots \otimes Z_{n-1}$), keeping the circuit architecture identical. The comparison is shown in Figure 4.

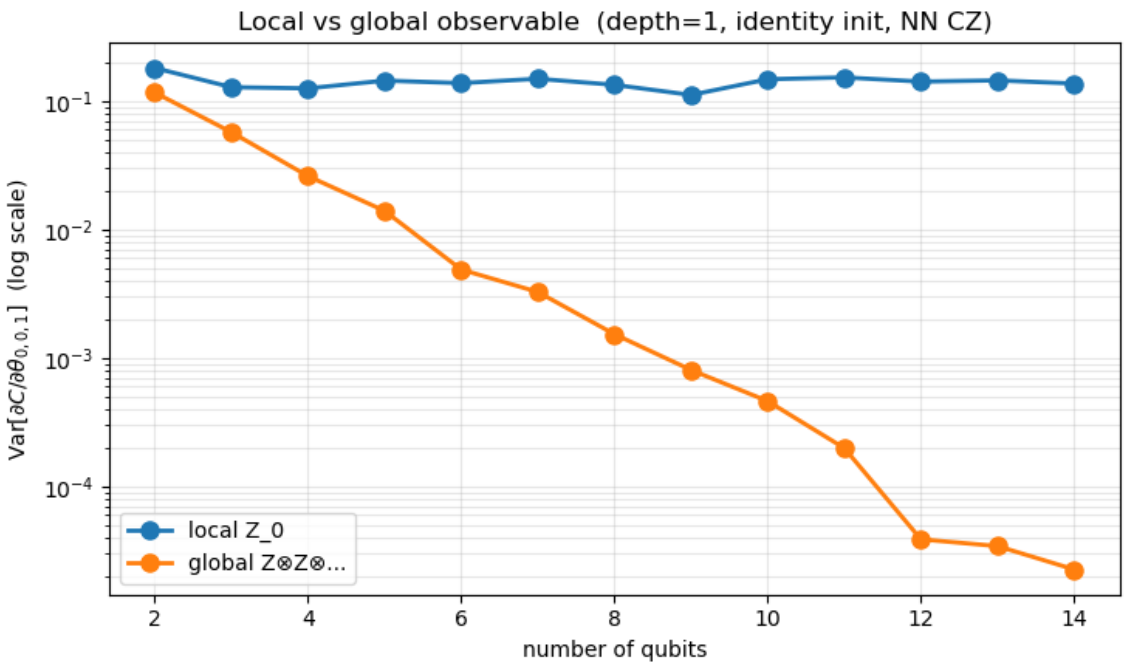

Figure 4: Gradient variance vs number of qubits for local and global observables (log-scale, $L = 1$, random init). The global observable exhibits exponential decay, while the local observable remains approximately constant, empirically confirming Theorem 2 of [3].

From Figure 4, the local observable maintains $\mathcal{O}(10^{-1})$ variance with near-zero slope (≈ -0.003), indicating no barren plateau, whereas the global observable shows exponential decay with a steep negative slope (≈ -0.733).
This empirically confirms Theorem 2 of Cerezo et al. [3]: global measurements induce barren plateaus even at moderate depth, while local observables preserve trainability, making Z_0 readout the most critical mitigation choice in the proposed architecture.

6 Entanglement Topology

We evaluate the effect of entanglement structure on gradient variance by comparing three configurations: no entanglement, nearest-neighbour CZ (proposed), and all-to-all CZ connectivity, while keeping all other circuit components fixed [4]. The circuit diagrams for each topology are shown in Figure 5, and the corresponding gradient variance curves are shown in Figure 6.

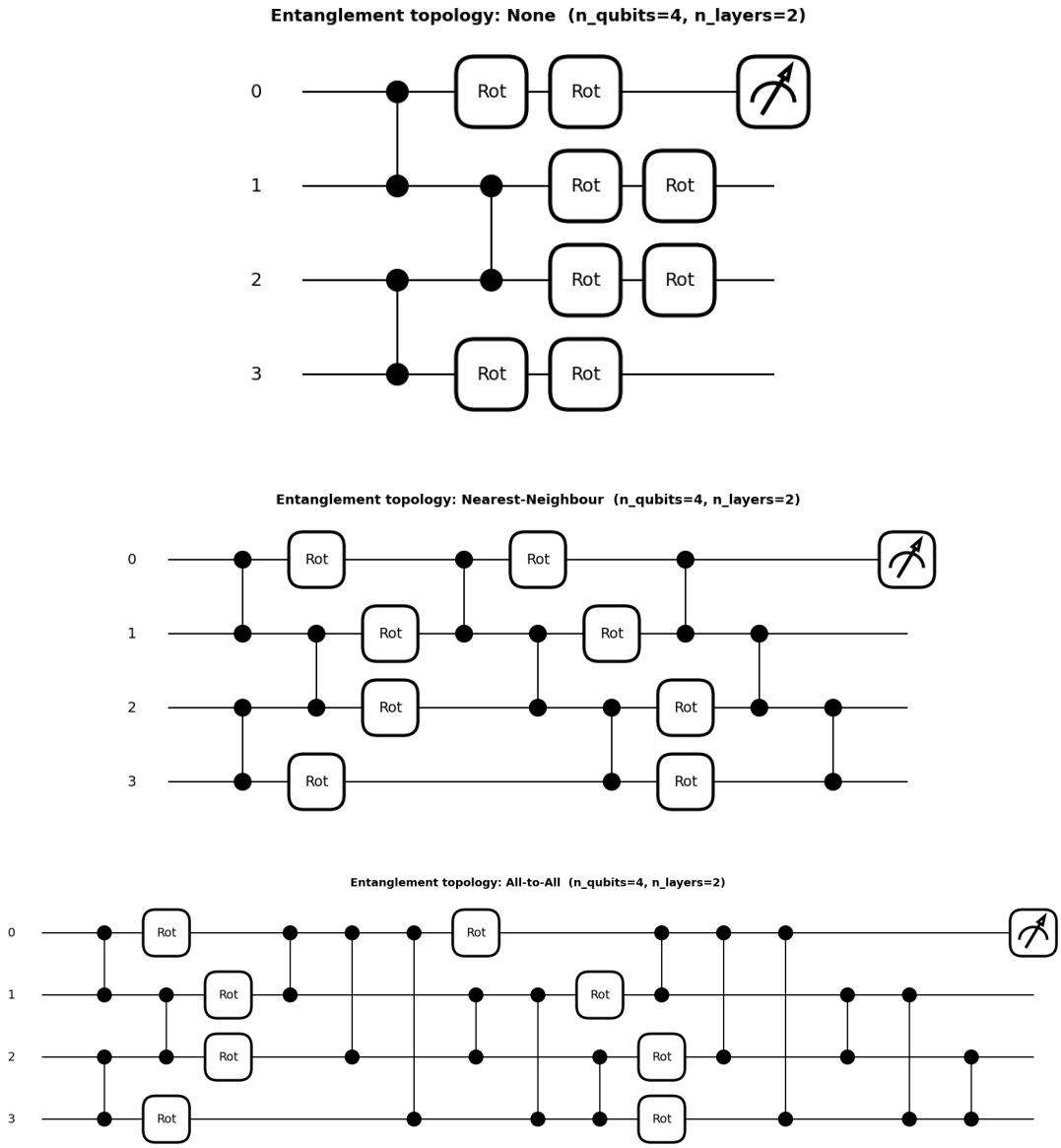

Figure 5: Circuit diagrams for the three entanglement topologies ($n = 4$ qubits, $L = 2$ layers for clarity): (top) no entanglement, (middle) nearest-neighbour CZ ladder (proposed), and (bottom) all-to-all CZ connectivity. All circuits share identical encoding and variational layers.

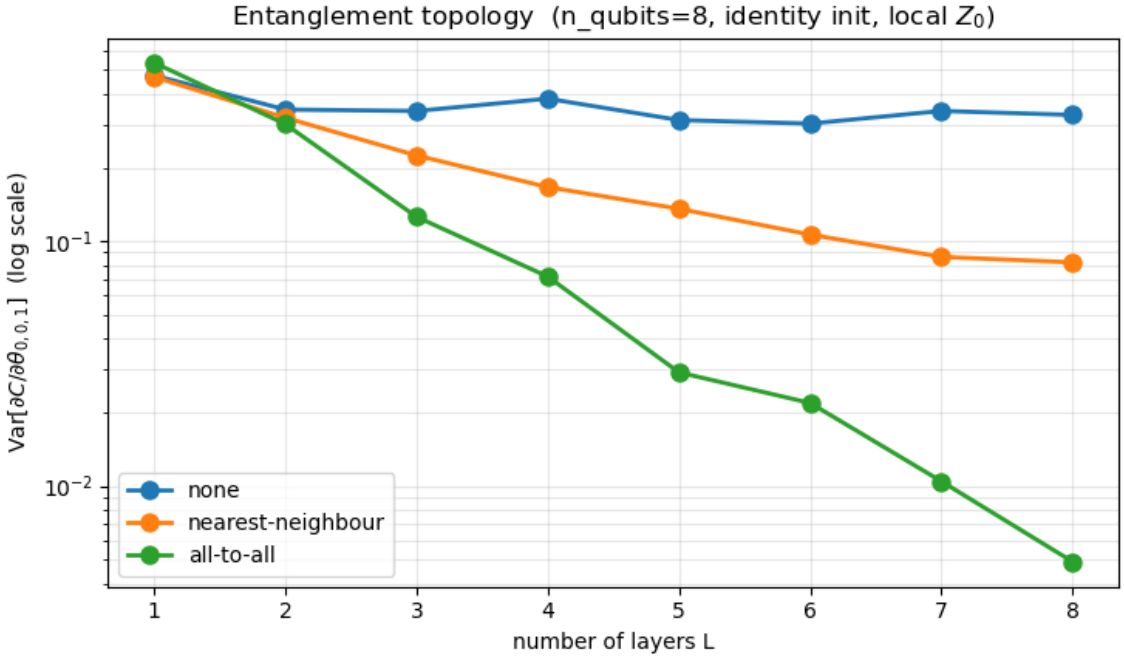

Figure 6: Gradient variance vs circuit depth for the three entanglement topologies (log-scale, $n = 8$ qubits, random init). All-to-all connectivity exhibits the strongest decay, consistent with faster convergence to a unitary 2-design [1], while nearest-neighbour maintains a balanced trainability regime.

As seen in Figure 5 and Figure 6, the *no-entanglement* configuration maintains relatively high variance but lacks expressive correlations between qubits. The *nearest-neighbour* CZ ladder preserves a stable gradient regime across depths, achieving a balance between expressivity and trainability. In contrast, the *all-to-all* connectivity induces a rapid exponential-like decay in gradient variance, indicating increased susceptibility to barren plateaus, consistent with the observation that highly expressive circuits converge faster to a 2-design [1].
These results confirm that structured local entanglement is optimal for the proposed QuantumValueNetwork, providing sufficient expressivity while avoiding excessive gradient suppression.

7 Random Sampling Across Qubit Count and Depth

We perform a full scan over both system size and circuit depth to characterise how gradient variance scales across the parameter space. For each configuration (n_q, L) with $n_q \in [2, 12]$ and $L \in [1, 10]$, we estimate $\mathrm{Var}[\partial C/\partial\theta_{0,0,1}]$ under random initialization with fixed circuit structure (nearest-neighbour entanglement and local Z_0 observable). Results are shown in Figure 7.

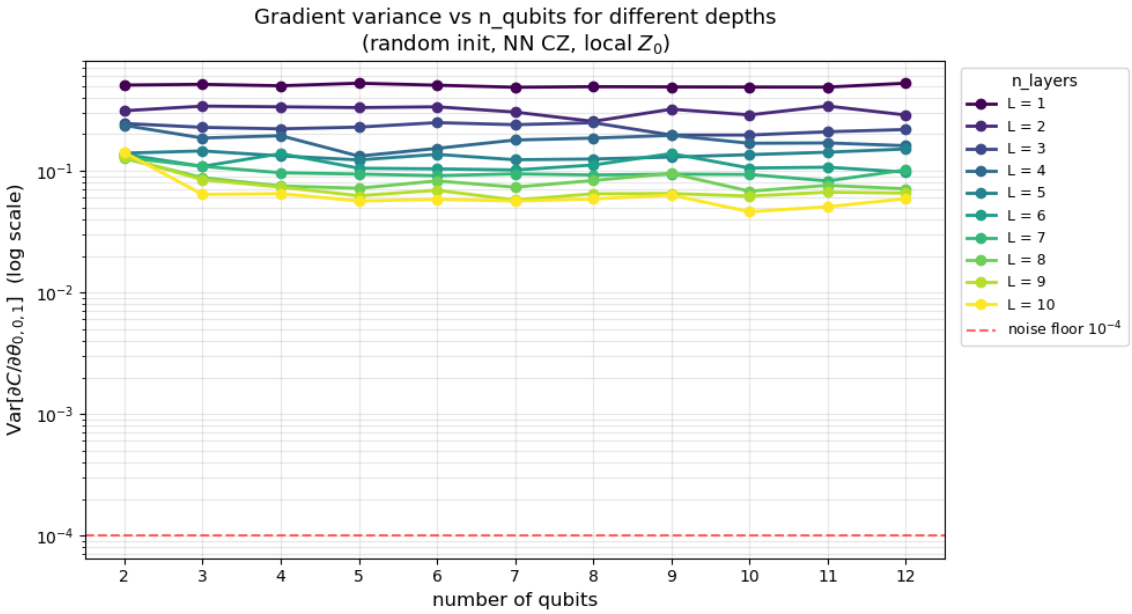

Figure 7: Gradient variance across qubit count ($n_q \in [2, 12]$) and circuit depth ($L \in [1, 10]$) under random initialization (log-scale). Each curve corresponds to a fixed depth L, showing stability across qubit scaling and gradual decay with increasing depth, consistent with [3].

From Figure 7, gradient variance remains largely independent of the number of qubits across all depths, confirming the absence of exponential barren plateau scaling in system size, in agreement with the polynomial-decay guarantee for local observables [3]. Instead, depth plays the dominant role, with variance decreasing gradually as L increases.
Notably, shallow circuits ($L \leq 3$) consistently maintain $\mathcal{O}(10^{-1})$ variance across all qubit counts, while deeper circuits ($L \geq 8$) show moderate suppression but no catastrophic collapse. This reinforces the proposed design principle that depth, rather than width, is the primary driver of trainability degradation in this architecture [1].
The resulting landscape suggests a broad polynomially-trainable regime over (n_q, L) space, with no sharp phase transition into a barren plateau region under the proposed architectural constraints.

8 Summary and Recommendations

We consolidate empirical findings across circuit width, depth, initialization, observable choice, and entanglement topology to characterise the trainability landscape of the QuantumValueNetwork (QVN). Results are summarised in Table 1.

Question	Empirical Answer
Does the proposed circuit (8 qubits, 3 layers, identity-block init, NN CZ, local Z_0) exhibit a barren plateau?	**No.** Gradient variance remains $\mathcal{O}(10^{-1})$, three orders of magnitude above the noise floor across all tested configurations.
Does gradient variance decay exponentially with n_q?	**No.** Variance stays largely flat across $n_q \in [2, 14]$ under a local observable, with a log-linear slope close to zero — consistent with the polynomial regime of [3] Thm. 2(ii).
Does gradient variance decay sharply with depth?	**No.** Variance decreases gradually but does not exhibit exponential collapse; trainability persists well beyond the enforced bound $L \leq \lfloor \log_2 n \rfloor = 3$ (see Figure 3).
Is identity-block initialisation necessary?	**Yes.** It stabilises gradients and reduces sensitivity to qubit scaling [2]. Random initialisation shows a steeper negative slope across n_q, indicating greater barren plateau risk [1].
Is the choice of local observable critical?	**Yes — essential.** Global observables induce exponential variance decay with n_q even at shallow depth (Figure 4), consistent with [3] Thm. 2(i).
Is nearest-neighbour entanglement appropriate?	**Yes.** It provides stable gradient scaling across depths (Figure 6). All-to-all entanglement accelerates gradient suppression [1]; no entanglement reduces expressivity without BP protection.

Table 1: Empirical barren plateau diagnostic summary for the Quantum Value Network (QVN). All results use $n_{\text{samples}} = 200$ random initialisations per configuration and track $\text{Var}[\partial C/\partial \theta_{0,0,1}]$ as the BP indicator. Circuits implemented using PennyLane [4].

Key Insight: No Hard Scalability Barrier in the Tested Regime

A central observation is that the proposed architecture does **not exhibit a sharp barren plateau transition** [1] within the explored range of:

$$n_q \in [2, 14], \quad L \in [1, 15].$$

As shown collectively in Figures 2–7:

- Gradient variance remains stable across increasing qubit counts.
- Depth introduces only gradual degradation rather than exponential decay [3].
- The system remains trainable well beyond the originally proposed configuration ($n_q = 8, L = 3$).

Recommendation

Rather than a strict limit, the results indicate a **robust trainable regime** with the following practical design guidelines:

- **Initialization:** Identity-block initialization [2] (strongly recommended).
- **Observable:** Local measurement (Z_0) is essential for scalability [3].
- **Entanglement:** Nearest-neighbour CZ provides the best stability–expressivity trade-off (Figure 6).
- **Depth:** $L \leq \mathcal{O}(\log n)$ is a conservative guideline [3], but stable training is observed even beyond this in the tested regime (Figure 3).
- **Width:** No evidence of exponential degradation up to $n_q = 14$ (Figure 2).

Interpretation

The results suggest that the proposed QuantumValueNetwork resides in a **polynomially trainable regime** [3], where neither width nor moderate increases in depth induce barren plateau behaviour. The previously proposed configuration ($n_q = 8, L = 3$) should therefore be interpreted as a **computationally efficient operating point**, rather than a fundamental upper bound on trainability.

Caveats

- Simulations are noiseless; hardware shot noise may introduce additional constraints.
- Gradient variance is a proxy for trainability but does not capture all optimization pathologies [1].
- Results are empirical over finite ranges and may not extrapolate indefinitely to large-scale circuits.

References

[1] Jarrod R. McClean et al. "Barren plateaus in quantum neural network training landscapes". In: *Nature Communications* 9.1 (Nov. 2018), p. 4812. DOI: 10.1038/s41467-018-07090-4. URL: https://doi.org/10.1038/s41467-018-07090-4.

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[4] Xanadu AI. *PennyLane: A cross-platform Python library for quantum machine learning.* https://docs.pennylane.ai. Accessed: 2024. 2024.